Heng Guo

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Homepage

Experience

Specially-Appointed Researcher — Beiing, China
School of Artificial Intelligence, Beijing University of Posts and Telecommunications — *Nov. 2023 – present*

Specially-Appointed Assistant Professor — Osaka, Japan
Osaka University — *May. 2022 – Oct. 2023*

Research Assistant — Osaka, Japan
Osaka University — *Oct. 2018 – Apr. 2022*

Research Intern — Yokohama, Japan
Japan Institute of Oppo Mobile Telecommunications Corp., Ltd — *June 2020 – Oct. 2020*

Research Intern — Beijing, China
AI Lab of Qihoo 360 Technology Corp., Ltd. — *June 2016 – Jan. 2017*

Education

University of Electronic Science and Technology of China (UESTC) — Chengdu, China
Bachelor of Electronic Engineer — *Aug. 2015 – Sep. 2011*

University of Electronic Science and Technology of China (UESTC) — Chengdu, China
Master of Signal Processing | Supervisors: Bing Zeng, Shuaicheng Liu — *Aug. 2018 – Sep. 2015*

Osaka University — Osaka, Japan
Doctor of Computer Science | Supervisors: Yasuyuki Matsushita, Boxin Shi — *Present – Sep. 2018*

Fundings

NSFC Excellent Young Scientists Fund (Overseas), **Pi**	2024 – 2026
NSFC General Program, **Pi**	2024 – 2027
NSFC Joint Fund for Enterprise Innovation and Development, **Participant**	2024 – 2027
Beijing−Tianjin−Hebei Basic Research Program, **Pi**	2024 – 2027
Hebei Provincial Major Science and Technology Support Program, **Pi**	2024 – 2026
Beijing Municipal International (Hong Kong, Macao, and Taiwan) Joint R&D Program, **Pi**	2025 – 2027

Publication

(∗ Equal contribution, † Corresponding author.)

[1] Yufei Han, Chu Zhou, Youwei Lyu, Qi Chen, Si Li, Boxin Shi, Yunpeng Jia, **Heng Guo**†, Zhanyu Ma. "PolGS++: Physically-Guided Polarimetric Gaussian Splatting for Fast Reflective Surface Reconstruction." International Journal of Computer Vision. (**IJCV 2026**).

[2] Ruiyang Chen, Feiran Li, **Heng Guo**†, Zhanyu Ma. "FlashNormal: Detailed Surface Normal Estimation from Flash and No-Flash Images." IEEE Transactions on Circuits and Systems for Video Technology. (**TCSVT 2026**).

[3] Zonglin Li, Kexian Xie, Feiran Li, Chu Zhou, Yixing Huang, Ruiyang Chen, Zhanyu Ma, **Heng Guo**†. "AD-GS: Anatomical Density-guided 3D Gaussian Splatting for Sparse-View CBCT Reconstruction." British Machine Vision Conference. (**BMVC 2026**).

[4] Tongyu Yang, **Heng Guo**, Xingyou Lin, Yasuyuki Matsushita, Xin Fan. "Uncertainty-guided Neural Radiance Fields against Diverse Degradation Factors." (**Neurocomputing 2026**).

[5] Tengjiao Yin, Jinglei Shi, **Heng Guo**, Xi Wang. "VIGOR: Video Geometry-Oriented Reward for Temporal Generative Alignment." European Conference on Computer Vision. (**ECCV 2026**).

[6] Jilong Zhang, Songyun Yang, Yufei Han, **Heng Guo**†, Zhanyu Ma, "Photometric Stereo Prior Boosted Sparse Multiview Stereo." IEEE International Conference on Image Processing. (**ICIP 2026**).

[7] Tongyu Yang, **Heng Guo**, Yasuyuki Matsushita, Fumio Okura, Yu Luo, Xin Fan. "High-fidelity Multi-view Normal Integration with Scale-encoded Neural Surface Representation." Computer Graphics Forum. (**CGF 2026**).

[8] Zaiyan Yang, Xinpeng Liu, **Heng Guo**†, Jinglei Shi, Zhanyu Ma, Fumio Okura. "NRGS: Neural Regularization for Robust 3D Semantic Gaussian Splatting." IEEE International Conference on Multimedia and Expo. (**ICME 2026**).

[9] Zonglin Li, Jieji Ren, Shuangfan Zhou, **Heng Guo**†, Jinnuo Zhang, Jiang Zhou, Boxin Shi, Zhanyu Ma, Guoying Gu. "Near-Light Color Photometric Stereo for Mono-Chromaticity Non-Lambertian Surface." IEEE International Conference on Acoustics, Speech and Signal Processing. (**ICASSP 2026**).

[10] Jinyi Chang, Dongliang Chang, Lei Chen, Bingyao Yu, **Heng Guo**, Zhanyu Ma. "Towards Privacy-Preserving Fine-Grained Visual Classification via Hierarchical Learning from Label Proportions." IEEE International Conference on Acoustics, Speech and Signal Processing. (**ICASSP 2026**).

[11] Wenjie Li, **Heng Guo**†, Yuefeng Hou, Zhanyu Ma. "FourierSR: A Fourier Token-based Plugin for Efficient Image Super-Resolution." IEEE Transactions on Image Processing. (**TIP 2026**).

[12] Wenjie Li, Jinglei Shi, Jin Han, **Heng Guo**†, Zhanyu Ma. "Seeing Through the Rain: Resolving High-Frequency Conflicts in Deraining and Super-Resolution via Diffusion Guidance." AAAI Conference on Artificial Intelligence. (**AAAI 2026**).

[13] Zhenyu Jin, Wenjie Li, Zhanyu Ma, **Heng Guo**†. "SpecGen: Neural Spectral BRDF Generation via Spectral-Spatial Tri-plane Aggregation." IEEE Winter Conference on Applications of Computer Vision. (**WACV 2026**).

[14] Wenjie Li, Xiangyi Wang, **Heng Guo**†, Guangwei Gao, Zhanyu Ma. "Self-Supervised Selective-Guided Diffusion Model for Old-Photo Face Restoration." Annual Conference on Neural Information Processing Systems. (**NeurIPS 2025**).

[15] Wenjie Li, **Heng Guo**†, Yuefeng Hou, Guangwei Gao, Zhanyu Ma. "Dual-domain Modulation Network for Lightweight Image Super-Resolution." IEEE Transactions on Multimedia. (**TMM 2025**).

[16] Yufei Han, Bowen Tie, **Heng Guo**†, Youwei Lyu, Si Li†, Boxin Shi, Yunpeng Jia, Zhanyu Ma. "PolGS: Polarimetric Gaussian Splatting for Fast Reflective Surface Reconstruction." IEEE International Conference on Computer Vision. (**ICCV 2025**).

[17] Kailong Zhang, Youwei Lyu, **Heng Guo**†, Si Li, Zhanyu Ma, Boxin Shi. "PolarAnything: Diffusion-based Polarimetric Image Synthesis." IEEE International Conference on Computer Vision. (**ICCV 2025 Highlight**).

[18] Chu Zhou, Yixin Yang, Junda Liao, **Heng Guo**, Boxin Shi, Imari Sato. "Polarimetric Neural Field with Unified Complex-Valued Wavefunction." IEEE International Conference on Computer Vision. (**ICCV 2025**).

[19] Mingzhi Pei, Xu Cao, Xiangyi Wang, **Heng Guo**†, Zhanyu Ma. "PMNI: Pose-free Multi-view Normal Integration for Reflective and Textureless Surface Reconstruction." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[20] Shuangfan Zhou, Chu Zhou, Youwei Lyu, **Heng Guo**†, Zhanyu Ma, Boxin Shi, Imari Sato. "PIDSR: Complementary Polarized Image Demosaicing and Super-Resolution." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[21] Mengqiu Xu, Kaixin Chen, **Heng Guo**, Yixiang Huang, Ming Wu, Zhenwei Shi, Chuang Zhang, Jun Guo. "MFogHub: Bridging Multi-Regional and Multi-Satellite Data for Global Marine Fog Detection and Forecasting." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2025**).

[22] Songyun Yang, Yufei Han, Jilong Zhang, Kongming Liang, Peng Yu, Zhaowei Qu, **Heng Guo**. "RotatedMVPS: Multi-view Photometric Stereo with Rotated Natural Light." IEEE Conference on Multimedia and Expo. (**ICME 2025 Oral**).

[23] Youwei Lyu*, **Heng Guo***, Kailong Zhang, Si Li, Boxin Shi. "SfPUEL: Shape from Polarization under Unknown Environment." Annual Conference on Neural Information Processing Systems. (**NeurIPS 2024**).

[24] Li Wenjie, **Heng Guo**†, Xuannan Liu, Kongming Liang, Jiani Hu, Zhanyu Ma, Jun Guo. "Efficient Face Super-Resolution via Wavelet-Based Feature Enhancement Network." ACM International Conference on Multimedia. (**ACM MM 2024**).

[25] **Heng Guo***, Jieji Ren*, Feishi Wang*, Boxin Shi, Mingjun Ren, Yasuyuki Matsushita. "DiLiGenRT: A Photometric Stereo Dataset with Quantified Roughness and Translucency." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2024**).

[26] Yufei Han, **Heng Guo**†, Koki Fukai, Hiroaki Santo, Boxin Shi, Fumio Okura, Zhanyu Ma, Yunpeng Jia. "NeRSP: Neural 3D Reconstruction for Reflective Objects with Sparse Polarized Images." IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2024**).

[27] Feishi Wang*, Jieji Ren*, **Heng Guo***, Mingjun Ren, Boxin Shi. "DiLiGenT-Pi: Photometric Stereo for Planar Surfaces with Rich Details –Benchmark Dataset and Beyond." International Conference on Computer Vision. (**ICCV 2023**).

[28] Jun Hoong Chan, Bohan Yu, **Heng Guo**, Jieji Ren, Zongqing Lu, Boxin Shi. "ReLeaPS: Reinforcement Learning-based Illumination Planning for Generalized Photometric Stereo." International Conference on Computer Vision. (**ICCV 2023**).

[29] Jipeng Lv, **Heng Guo**†, Guanying Chen, Jinxiu Liang, Boxin Shi. "Non-Lambertian Multispectral Photometric Stereo via Spectral Reflectance Decomposition." International Joint Conferences on Artificial Intelligence (**IJCAI 2023**).

[30] **Heng Guo**, Boxin Shi, Yasuyuki Matsushita. "Neural BRDF Plugin for Unsupervised Photometric Stereo." IEEE International Conference on Network Intelligence and Digital Content (**IC-NIDC 2023 best paper**).

[31] Feiran Li*, **Heng Guo***, Hiroaki Santo, Fumio Okura, Yasuyuki Matsushita. "Learning to Synthesize Photorealistic Dual-pixel Images from RGBD frames." International Conference on Computational Photography (**ICCP 2023**).

[32] Lilika Makabe, **Heng Guo**, Hiroaki Santo, Fumio Okura, Yasuyuki Matsushita. "Near-light Photometric Stereo with Symmetric Lights." International Conference on Computational Photography. (**ICCP 2023**).

[33] **Heng Guo**, Fumio Okura, Boxin Shi, Takuya Funatomi, Yasuhiro Mukaigawa, Yasuyuki Matsushita. "Multispectral Photometric Stereo for Spatially-Varying Spectral Reflectances." International Journal of Computer Vision. (**IJCV 2022**).

[34] **Heng Guo***, Zhipeng Mo*, Boxin Shi, Feng Lu, Sai-Kit Yeung, Ping Tan, Yasuyuki Matsushita. "Patch-based Uncalibrated Photometric Stereo under Natural Illumination." IEEE Transactions on Pattern Analysis and Machine Intelligence. (**TPAMI 2021**).

[35] **Heng Guo**, Fumio Okura, Boxin Shi, Takuya Funatomi, Yasuhiro Mukaigawa, Yasuyuki Matsushita. "Multispectral Photometric Stereo for Spatially-Varying Spectral Reflectances: A Well-posed Problem?" IEEE Conference on Computer Vision and Pattern Recognition. (**CVPR 2021**).

[36] **Heng Guo**, Boxin Shi, Michael Waechter, Yasuyuki Matsushita. "Self-calibrating Near-light Photometric Stereo under Anisotropic Light Emission." Meeting on Image Recognition and Understanding (**MIRU 2020 outstanding paper**).

[37] **Heng Guo**, Shuaicheng Liu, Shuyuan Zhu, Heng Tao Shen, Bing Zeng. "View-Consistent MeshFlow for Stereoscopic Video Stabilization." IEEE Transactions on Computational Imaging. (**TCI 2018**).

[38] **Heng Guo**, Shuaicheng Liu, Tong He, Shuyuan Zhu, Bing Zeng, Moncef Gabbouj. "Joint Video Stitching and Stabilization from Moving Cameras." IEEE Transactions on Image Processing (**TIP 2016**).

[39] **Heng Guo**, Shuaicheng Liu, Shuyuan Zhu, Bing Zeng. "Joint Bundled Camera Paths for Stereoscopic Video Stabilization." International Conference on Image Processing. (**ICIP 2016**) (Oral).

Granted Patents

[1] **Heng Guo**, Zhanyu Ma, Boxin Shi, Kongming Liang, Yunpeng Jia, Yufei Han. "3D Reconstruction Method, Apparatus, Device, and Storage Medium." Chinese invention patent, ZL202410352394.0, granted October 9, 2026.

[2] **Heng Guo**, Zhanyu Ma, Ruiyang Chen, Zaiyan Yang. "A Surface Normal Estimation Method, Apparatus, Device, and Medium for Target Objects." Chinese invention patent, ZL202510094535.8, granted June 19, 2026.

[3] **Heng Guo**, Zhanyu Ma, Mingzhi Pei, Xiangyi Wang. "A 3D Reconstruction Method, Apparatus, Device, and Medium Based on Multi-view Normal Aggregation." Chinese invention patent, ZL202510110808.3, granted April 3, 2026.

[4] **Heng Guo**, Wenjie Li, Zhanyu Ma. "Methods for Training an Image Reconstruction Model and Reconstructing Images." Chinese invention patent, ZL202510156115.8, granted December 2, 2025.

Academic Services

- **Committee Member**: China Institute of Electronics Virtual Reality Branch
- **Guest Editor**: Journal of the Optical Society of America (JOSA)
- **Journal Reviewer**: IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI), International Journal of Computer Vision (IJCV)
- **Conference Reviewer**: CVPR, ECCV, ICCV, NeurIPS, AAAI
- **Demo Chair**: ACCV 2026
- **Forum Chair**: The 24th CCF-CV "Vision Without Limits" Seminar Series
- **Organizing Chair**: Intelligent Scientific Imaging Forum, CSIG Young Scientists Conference 2025
- **Organizing Chair**: CAA Conference on Artificial Intelligence and Robotics Education 2025
- **Invited Speaker**: The 22nd Forum on Information Technology (FIT), Japan

Honors

Outstanding Reviewer, CVPR 2026	Oct. 2026
First Prize, 6th Beijing Teaching Innovation Contest for College Teachers	Apr. 2026
Third Prize, 6th National Teaching Innovation Contest for College Teachers	July 2026
Best Paper, International Conference on Network Infrastructure and Digital Content (Top 1%)	Oct. 2023
Best Student Paper, Meeting on Image Recognition and Understanding (Top 1%)	Apr. 2020
First Prize, Beijing Education and Teaching Achievement Award, "Reform and Innovation of General AI Education for First-Year Students"	May 2026
Second Prize, Beijing College Student "AI+" Innovation Competition, "Reviving Heritage: Bringing Artifacts to Life with AI" (Faculty Advisor)	Jan. 2026
2025 High-Quality Undergraduate Courseware Award for Beijing Universities, "Introduction to Artificial Intelligence"	Mar. 2026
Outstanding Bachelor's Thesis Advisor, Beijing University of Posts and Telecommunications	June 2026
Excellent Master Thesis of UESTC (Top 3%)	June 2018
Excellent Postgraduate of UESTC (Top 6%)	June 2018
National Scholarship of China (Top 2%)	Oct. 2017